Zero-free quark determinants and the strong-CP angle in a modular flavour setup: an interpretation, a test, and two questions

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Summary. The setup below is modular-invariant, has CP imposed, and has a Dedekind-η counterterm. In it, $\bar\theta$ is constant over the whole upper half-plane exactly when no quark is massless at any interior point τ. Ref. [1] states this condition as an assumption. We read it as a principle and tested its consequence for the leptons at the values of τ fixed by the quark data. That test is negative: nothing fits beyond chance. One feature of the setup, a multiplier that does not cancel, leads to the first of two questions.

1. Setup

The three generations of each quark field form one triplet ρ of the metaplectic cover of $SL_2(\mathbb{Z})$. It is realised on $u=(\theta_2^2,\theta_3^2,\theta_4^2)$ and has determinant character $\det\rho=\chi_\eta^9$, where χ_η is the multiplier of η. Each field transforms as ρ or $\bar\rho$ ($s_f=\pm1$).

A Yukawa coupling of type (s_L,s_R) and weight k is $Y(\tau)=\eta^{-m}W(u)$. Here W is a 3×3 matrix of polynomials with $W(A_\gamma u)=A_L W A_R^\top$ for $\gamma=S,T$, where A_γ is the multiplier of u, and A_L,A_R equal $\bar A_\gamma$ or A_γ for a field in ρ or $\bar\rho$. The exponent is $m=18,6,0$ for $\rho\otimes\rho$, $\bar\rho\otimes\bar\rho$ and the mixed types.

CP is imposed, so the coefficients are real in a real basis and $Y(-\bar\tau)=\overline{Y(\tau)}$. In our sign convention, $\bar\theta=\theta_0-\arg\det(Y_uY_d)+N\arg\eta(\tau)$, with a counterterm $N=6K$, where $K=k_u+k_d$. The setup is tree level, with holomorphic couplings and no supersymmetry, so loop corrections to $\bar\theta$ are outside its scope. Its ingredients are stated as hypotheses, not derived.

The quark data are running masses at M_Z [6] and the PDG 2025 $|V_{ij}|$ [7]. Each observable is fitted as a logarithm, with its relative error combined in quadrature with a 10% theory allowance; J is not in the χ^2.

2. The lemma, read physically

The character of $\det\rho$ gives $\det Y=\eta^a g$, with $a=6,18,0$ by type and g a level-one form of weight $w=3k-a/2$. With real coefficients and $N=6K$, $\bar\theta$ is constant on $\mathbb{H}$ iff $g_ug_d/\eta^{2(w_u+w_d)}$ is constant. By the valence formula, and because j maps onto $\mathbb{C}$, that holds iff each g is zero-free on $\mathbb{H}$, that is iff $g=c\,\Delta^{w/12}$. Physically: $\bar\theta$ is constant over the moduli space exactly when no quark is massless at any interior τ. Checked on all sixteen (type, $k\le7$) at 50 digits:

class	w	$\det Y$	massless point in $\mathbb{H}$
automatic	0	$c\,\eta^{6k}$	none
tuned, generic	12	$\eta^a(\alpha E_4^3+\beta\Delta)$	one, at $j(\tau_0)=-\beta/\alpha$; on a CP locus, since $\alpha,\beta\in\mathbb{R}$
tuned, on the variety $\alpha=0$	12	$c\,\eta^{6k}$	none: one real condition moves the zero to the cusp
impossible	6, 18	$\eta^a E_6\times(\dots)$	at $\tau=i$, for every coupling
identically zero	-6	0	($\bar\rho\otimes\bar\rho$ at $k=1$)

3. The multiplier

For zero-free couplings, $\det(Y_uY_d)=c\,\eta^{6K}$. In this setup the counterterm cancels the QCD anomaly of the modular transformations only if $6K\equiv-9(2s_Q+s_u+s_d)\pmod{24}$, and for zero-free couplings that holds automatically.

Its right side vanishes modulo 24 only for vectorial hands ($s_u=s_d=-s_Q$). Those assignments fail the quark fit, with $\chi^2=47.9$ and 54.6 for 2 dof. Up to weight 7, no assignment with a degree of freedom fits; the best found is $\chi^2=47.9$. The four that are compatible with the data are not vectorial: $K=11,11,10,14$. So $24\nmid6K$, and η^{6K} carries the multiplier $e^{i\pi K/2}$ under $\tau\to\tau+1$. Ref. [1] works with Δ^m and avoids multipliers (its footnotes 4 and 6, and §2.4).

4. An interpretation

We read the zero-free condition as a principle: the holomorphic invariants of the couplings, here the mass determinants, vanish only at the cusp $\tau\to i\infty$. This restates ref. [1]'s assumption as a principle; it does not derive it. Refs. [2,3,8] instead reach a constant determinant by cancelling weights. Given CP and the counterterm, the principle fixes $\bar\theta\in\{0,\pi\}$ at every τ. It excludes mechanisms that rest on a mass matrix of reduced rank at an interior fixed point, such as the rank-one charged-lepton matrix at $\tau=i$ of ref. [9]; ref. [1]'s assumption does the same for the quarks. Because $\bar\theta\approx0$ was known beforehand, the lepton test below is the principle's first test with an outcome not known in advance.

5. The lepton test

The rule, the code and the output were each committed before being read (minor exceptions are disclosed in the repository). Checks marked *post hoc* were added afterwards.

Three of the four compatible quark assignments ($K=11,11,10$) have as many parameters as observables, so their solutions fix τ. The fourth ($K=14$) has two more parameters and is left out. That leaves 36 distinct values of τ, counted modulo $SL_2(\mathbb{Z})$ and $\tau\to-\bar\tau$: 16 exact fits, and 20 compatible at $\chi^2\le3.84$.

At each τ the same zero-free condition is imposed on the charged-lepton coupling (s_L,s_e). It is imposed too on the Weinberg coupling, which we take in the symmetric part of the (s_L,s_L) space; that choice, with no right-handed neutrinos, is ours. Ten

coupling pairs are admissible, with 2 to 6 real parameters at fixed τ. They face six observables: m_e/m_μ and m_μ/m_τ [6], and $\sin^2\theta_{12,13,23}$ and $\Delta m^2_{21}/|\Delta m^2_{3\ell}|$ from NuFIT 6.0 with Super-Kamiokande atmospheric data [5]. Nine of the pairs have a degree of freedom, giving $9 \times 36 = 324$ tests.

8 of the 324 tests pass at 95%, with the 10% allowance. The post hoc checks then remove or weaken seven of them:

- four sit at compatible-only quark points where $|J_{\rm CKM}|$ is 5.5 to 5×10^4 times too small;
- one gives $\Sigma m = 0.13$ eV, above current cosmological bounds;
- two are weakened once J is added to the quark χ^2.

One survives, at an exact quark point. It has normal ordering, $\Sigma m \simeq 0.064$ eV, $m_{\beta\beta} \simeq 4$ meV and $\sin^2\theta_{23} = 0.431$. That value is just below NuFIT 6.0's 3σ range with Super-Kamiokande data (0.435–0.585), and inside it without them.

A family-wide null (*post hoc*: 40 random lepton targets, each fitted over all 324 tests) gives $p = 0.125$. Nothing fits beyond chance. Ref. [1] likewise finds that its minimal lepton choice does not fit at the quark-fixed τ, and it adds two parameters. The two results are of different kinds: theirs concerns one model, ours a scan with a look-elsewhere null.

6. Two questions

1. **The multiplier.** In the notation of ref. [1], our counterterm is its eq. (24) with $k_{f_3} = -3K$, so $e^{-8\pi^2 f_3} \propto \eta^{-6K}$.

   - Ref. [1] sets η's multiplier aside because only η^{24} is needed (footnote 4). It notes that Yukawa multipliers possibly cancel in $\det Y$ (footnote 6), and it chooses its lepton structure to avoid multipliers (§2.4).
   - Here the multipliers do not cancel. Since $\det\rho = \chi_\eta^9$ and none of their hands is vectorial (§3), the four compatible assignments have $\det(Y_u Y_d) = c\,\eta^{6K}$ with $24 \nmid 6K$, so the QCD anomaly is cancelled only because the threshold term carries the same multiplier.

   Is such a term, with its multiplier matched by phases in the matter transformations, known to arise in a string completion? (Ref. [1], §3, allows such phases for $W \propto \eta^{-2k_W}$.) Or does consistency force the weight of $\det(Y_u Y_d)$ to be a multiple of 12, which would exclude these assignments?

2. **The principle.** Ref. [1] states the zero-free condition as an assumption, "perhaps realised in some string compactifications" (§2.1). Ref. [4] argues that the limit $Y \to 0$ lies at infinite distance along large complex-structure directions. It notes zeros at finite distance through D-brane position moduli, and sets them aside. For a theory with a single modulus τ and full $SL_2(\mathbb{Z})$, is any argument known that excludes an interior zero of $\det Y$, or is the question open?

Reproducibility. The rules, code and outputs are public, with the commit at which each was fixed: `https://github.com/originaxiom/origin-axiom/tree/7e01601a8/relays/chat1_2026-10-21_zero_free`. No Standard Model parameter is claimed as derived.

[1] F. Feruglio, A. Marrone, A. Strumia, A. Titov, arXiv:2505.20395. [2] F. Feruglio, A. Strumia, A. Titov, JHEP 07 (2023) 027, arXiv:2305.08908. [3] F. Feruglio, M. Parriciatu, A. Strumia, A. Titov, JHEP 08 (2024) 214, arXiv:2406.01689. [4] G. F. Casas, L. E. Ibáñez, F. Marchesano, arXiv:2403.09775. [5] I. Esteban *et al.* (NuFIT 6.0), JHEP 12 (2024) 216, arXiv:2410.05380. [6] G.-y. Huang, S. Zhou, Phys. Rev. D 103 (2021) 016010, arXiv:2009.04851. [7] S. Navas *et al.* (Particle Data Group), Phys. Rev. D 110 (2024) 030001, and 2025 update. [8] J. T. Penedo, S. T. Petcov, JHEP 10 (2024) 172, arXiv:2404.08032. [9] F. Feruglio, V. Gherardi, A. Romanino, A. Titov, JHEP 05 (2021) 242, arXiv:2101.08718.